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To cite this article: Diane R. Edmondson & Lucy M. Matthews (2020) Does Sleep Really Matter? Examining Sleep among Salespeople as Boundary Role Personnel for Key Job Factors, Journal of Business-to-Business Marketing, 27:1, 71-79, DOI: [10.1080/1051712X.2020.1713557](https://doi.org/10.1080/1051712X.2020.1713557)

To link to this article: <https://doi.org/10.1080/1051712X.2020.1713557>



Published online: 21 Jan 2020.



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Does Sleep Really Matter? Examining Sleep among Salespeople as Boundary Role Personnel for Key Job Factors

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ABSTRACT

Purpose: Sleep deprivation among employees has become commonplace in the workforce. In the United States, the number of hours individuals sleep per night has declined by an hour and a half per night since the 1960s. As of 2005, seventy-four percent of individuals were getting less than eight hours of sleep per night on weekdays. There are negative ramifications to the organization when employees are sleep deprived such as lost productivity, increased accident rate, and increased absenteeism.

Originality/Value/Contribution of the paper: Although prior research has investigated the impact that sleep deprivation has on unique job positions (e.g., shift workers, transportation), no research has investigated how sleep impacts business-to-business sales employees. Due to the responsibilities of business-to-business sales employees, this type of position should not just be lumped in with other organizational employees. For example, business-to-business sales employees are boundary spanning employees that are responsible for generating revenue for the organization. These sales employees are also more likely to be physically, socially, and psychologically separated from the organization since they frequently work outside of the office. Because of these distinctive roles, this study examines if differences based on sleep duration exist for business-to-business sales employees for two individual and five organizational factors. The two individual factors consisted of grit and happiness while the five organizational factors consisted of perceived organizational support, perceived supervisory support, job satisfaction, organizational commitment, and turnover intentions.

Methodology/Approach: Data was collected from 320 business-to-business sales employees using a cross industry survey. The survey was conducted using a Qualtrics panel. Sleep duration was broken into two groups with one group consisting of business-to-business sales employees who sleep, on average, seven or more hours per night and the other group consisting of business-to-business sales employees who sleep, on average, six or less hours per night. The sleep duration groups were determined based on prior research that found adults should regularly sleep seven or more hours per night in order to support optimal health. There were one hundred thirty-four respondents who slept six hours or less and one hundred eighty-six respondents who slept seven or more hours, on average, per night.

Findings: The research question developed was answered by examining the differences between means for the two individual constructs and the five key organizational constructs. Differences in sleep duration for six of the seven sales constructs. There were significant differences between those that slept seven or more hours per night from those that slept six or less for grit, perceived organizational support, perceived supervisory support, job satisfaction, organizational commitment, and turnover intentions. Results found that business-to-business sales employees that slept seven or more hours per night exhibited more grit than those that slept six or less hours per night. In addition, business-to-business sales employees that slept six or less hours per night had higher perceptions of both organizational and supervisory support than those that slept seven or more hours per night. Furthermore, business-to-business sales employees that slept six or less hours per night displayed higher job satisfaction and organizational commitment than sales employees that slept seven or more hours per night. Finally, business-to-business sales employees that slept seven or more hours per night indicated higher levels of turnover intentions than sales employees that slept six or less hours per night. There were no significant differences between the two sleep groups for sales employee happiness. Although some of these results may appear to be counter-intuitive, conservation of resources theory can be used to explain these findings.

Research Implications: This exploratory study showcases the importance of good sleep among business-to-business sales employees. This study draws on the conservation of resources theory literature to explain the counterintuitive findings. This theory posits that individuals will retain and guard against resource depletion. Additionally, as resources become depleted, other resources are valued to a greater extent. In this study, the resource being depleted is the condition of sleep.

KEYWORDS

Sales; sleep; grit; support; job satisfaction; organizational commitment; turnover intentions

When business-to-business sales employees experience a lower average amount of sleep per night (i.e., six hours or less), they perceive higher levels of support, job satisfaction, and organizational commitment than business-to-business sales employees who experience a higher average amount of sleep per night (i.e., seven hours or more). Although this study provides a good initial examination of how sleep impacts sales organizations, this study also highlights the need for additional business-to-business sales employee sleep research. This is the first study that has examined how sleep duration impacts business-to-business sales employees.

Practical Implications: This study is important to sales research because it highlights the impact that poor sleep of business-to-business sales employees has on the organization. For sales organizations, we provide guidance on ways a sales organization can help promote healthy sleep for their business-to-business sales employees. Some examples of guidance provided can be categorized into four primary categories. These categories include (1) proper sleep education, (2) sales management training, (3) creating a sleep friendly workplace environment, and (4) creating specific workplace policies to minimize sleep disturbances such as setting reasonable work hours and sleep accommodating travel schedules.

Introduction

Approximately *seventy five percent of adults* have experienced *at least one symptom* of sleep issues a few nights a week or more (Scott and Judge 2006). Although adequate sleep duration has been defined to be between seven and nine hours per night (Hirshkowitz, Whiton, and Albert et al. 2014), the average in the United States has declined from 8.5 hours per night in the 1960s to only seven hours a night by 2000 (Buxton et al. 2009). Furthermore, only 26% of individuals have been shown to sleep eight or more hours per night on weekdays as of 2005 (Scott and Judge 2006).

This shift in the amount of sleep obtained by individuals has *negative repercussions* on not only the individual but their employer as well. The socioeconomic costs of troubled sleep in the United States in 2011 is estimated to be between 63 and 91 billion dollars per year (Nielsen et al. 2018). More specifically, the direct and indirect health care costs among shift workers ranges between 30 and 40 billion dollars annually (Robbins and Jean-Louis 2018). Like shift workers, salespeople have unique job requirements such as out of town travel responsibilities and unusual work hours to meet with clients that set them apart from other employees. Although research has already shown the negative cost of sleep deprivation on businesses, since salespeople are truly the revenue generators for a firm, the impact of sleep deprivation on this employee type warrants additional investigation.

However, little to no research has investigated the *impact that sleep duration has on salespeople*. This exploratory study seeks to fill this gap. In essence, the goal of this study is to understand

how sleep deprivation, defined as sleeping, on average, less than seven hours per night, impacts key individual and organizational characteristics for business-to-business salespeople.

Literature review

Sleep duration

The benefits of obtaining adequate sleep has been well researched in many different disciplines at the individual level. For example, research has shown that sleep duration is linked to subjective well-being such as positive mood and happiness (Barber, Rupperecht, and Munz 2014; Norlander, Johansson, and Bood 2005). Research has also shown that those who sleep at least seven hours per night have the lowest risk of mortality (Buxton et al. 2009). Individuals obtaining adequate sleep also have a lower risk of weight gain, diabetes, hypertension, and cardiovascular disease (Buxton et al. 2009). In addition, individuals obtaining adequate sleep have higher memory performance and decreased forgetfulness (Abichou et al. 2019).

Research has also shown that a lack of sleep can have negative ramifications on organizational employees. Poor sleep has been shown to negatively impact businesses through lost productivity, increased accident rates, and reduced employee attendance (Gaultney and Collins-McNeil 2009). In addition, research has shown that poor sleep causes employees to have impaired cognition (Gaultney and Collins-McNeil 2009). This impaired cognition includes difficulty in decision making, especially of complex tasks (Harrison and Horne 1999; Scott and Judge 2006).

Sleep deprivation has been shown to negatively affect morale and social relationships as well as reduced physical and emotional health (Gaultney and Collins-McNeil 2009; Pilcher and Huffcutt 1996). Research has revealed that even short term sleep loss creates heightened emotional reactivity to even minor stressful events since sleep deprived individuals are too tired to cope effectively with stressors (Barber, Rupperecht, and Munz 2014). Physical health includes the increased feelings of fatigue and energy depletion due to poor sleep (Barber, Rupperecht, and Munz 2014). Caffeine consumption has been shown to not counteract the negative impacts of sleep deprivation (Gaultney and Collins-McNeil 2009).

Individual employee characteristics

In this study, sleep duration is investigated with two individual employee characteristics. These individual employee characteristics include grit and happiness. Grit is defined as “working strenuously toward challenges, maintaining effort and interest over years despite failure, adversity, and plateaus in progress” (Duckworth et al. 2007, p. 1087–1088). Findings reveal that grit is positively correlated with performance (Rodriguez et al. 2019). In the sales literature, grit is an important characteristic for hiring managers when looking for new sales employees (Rodriguez et al. 2019). However, no research was found that examines sleep duration and grit using a sales sample. In fact, only one Malaysian study was found which indicated that sleep quality impacted the grit level of undergraduate students (Siah et al. 2018).

Happiness is defined as a person “who frequently experiences positive emotions such as joy, happiness, and contentment” (Walsh, Boehm, and Lyubomirsky 2018, 200). In the literature, happiness is sometimes referred to as subjective well-being although this construct also includes negative emotions not found in happiness such as anger and sadness (Walsh, Boehm, and Lyubomirsky 2018). Prior research indicates that happy individuals typically experience more success than less happy individuals (Walsh, Boehm, and Lyubomirsky 2018); however, happiness is impacted by sleep. For example, Norlander, Johansson, and Bood (2005) found that healthy sleep patterns were linked to subjective well-being while Scott and Judge (2006) found that insomnia was associated with

decreased happiness. Additionally, it only takes one night of sleep deprivation to impair one’s emotions (Pilcher and Huffcutt 1996).

Organizational characteristics

Besides the two individual characteristics, the impact that sleep duration has on five organizational characteristics that are common in the sales literature are investigated in this study. These five organizational characteristics include perceived organizational support, perceived supervisory support, job satisfaction, organizational commitment, and turnover intentions. In the sleep literature, no research has investigated sleep duration of sales employees with any of these five organizational characteristics. A brief description of each of these organizational characteristics is as follows:

Perceived organizational support (POS) is defined as employees’ “beliefs concerning the extent to which the organization values their contribution and cares about their well-being” (Eisenberger et al. 1986, 501). Perceived supervisory support (PSS) examines the extent to which employees perceive their supervisor values them (Edmondson and Boyer 2013). Research has shown the both POS and PSS are positively related to organizational commitment and job satisfaction and negatively related to turnover intentions (Edmondson and Boyer 2013; Riggall, Edmondson, and Hansen 2009). However, greater supervisor support is significantly associated with adequate sleep in U.S. motor freight workers (Buxton et al. 2009). In addition, lower supervisory support reduced sleep quality in Latino farmworkers (Sandberg et al. 2016). For organizational support, low organizational support increased sleep disorders in Romanian University employees (Constantin and Maidaniuc-Chirila 2016).

Job satisfaction is defined as “a pleasurable or positive emotional state resulting from the appraisal of one’s job or job experiences” (Locke 1976, 1300). Research has shown that insomnia harms job satisfaction (e.g., Barnes, Miller, and Bostock 2017; Scott and Judge 2006). For sleep duration, Barnes, Ghumman, and Scott (2013) found that sleep had a positive relationship with job satisfaction. Organizational commitment examines an employee’s attachment and psychological identification to the organization (Jaramillo, Mulki, and Solomon

2006). Although no research exists which has examined organizational commitment and sleep duration, there have been calls for research on this topic in the literature (e.g., Barnes, Ghumman, and Scott 2013). Turnover intentions is defined as an employee's likelihood of ending their employment with a particular organization (Edmondson and Boyer 2013). Limited research exists examining sleep and turnover intentions. Although not the focus of the study, Tucker, Jimmieson, and Jamieson (2018) showed a positive correlation between sleep disturbances and turnover intentions for transport workers in Australia.

Business-to-business salespeople

Although *some differences in employee position type based on sleep have been investigated (e.g., shift workers, transportation), no research has been found that focuses on the business-to-business sales employee.* Sales employees are different from many other occupations due to the nature of their role. As a boundary spanning employee, sales employees typically face significant stress (Edmondson, Matthews, and Ambrose 2019) due to the fact that these employees engage "in job-related interactions with a person who is considered part of the environment, who is not a member of the organization" (Robertson 1995, 75). In fact, their success can hinge on the fact that the salesperson can recognize customer emotions and adapt accordingly (Kidwell et al. 2011; Siadou-Martin et al. 2017). These business-to-business sales employees are viewed as the revenue generators for their organization (Siguaw, Kimes, and Gassenheimer 2003); however, they are also typically physically, socially, and psychologically separated from the organization (Schwepker 2019; Mulki et al. 2008). This is because salespeople are more likely to be away from direct supervision since many work outside of the office (Bush et al. 2017; DeConnick 2011; Itani, Jaramillo, and Chonko 2019). They are often motivated by commission structures that focus on short-term goal achievements (Obal and Morgan 2018) and influenced by coercive, legitimate, and referent power of their sales managers (Mallin and Ragland 2017). Due to the unique nature of the sales position (e.g., Bush et al. 2017; Verbeke, Dietz, and Verwall 2011), it is likely that the negative ramifications of reduced sleep duration would be more exaggerated.

Because of this, the purpose of this paper is to answer the following research question:

RQ: Are there differences in key job factors including (a) grit, (b) happiness, (c) perceived organizational support, (d) perceived supervisory support, (e) job satisfaction, (f) organizational commitment, and (g) turnover intentions based on the level of sleep an employee gets on average?

Methodology

Business-to-business salespeople from across the United States were surveyed using a Qualtrics online panel. *Panel participants* worked in a variety of industries such as *wholesale/distribution and services (business, financial, consumer)* in order to increase generalizability. To help guarantee that respondents were actively participating in the survey, four verification checks were included in the survey instrument. Anyone that did not answer these verification checks correctly were pulled from the final dataset.

Three hundred twenty f (323) respondents successfully completed the entire survey. Forty-six percent of the respondents were male. Respondents ages ranged from 19 to 79 with an average age of 46 years of age. Fifty-seven percent of the respondents were married while sixty-seven percent did not have children under the age of 18 living in the home. Fifty-nine percent of the respondents had at least a bachelor's degree while 31% had some college and 10% had completed high school/GED. Thirty-five percent of respondents had an annual household income of \$50,000 or less, 43% reported a household income between \$50,000 and \$100,000, and 22% listed a household income exceeding \$100,000. The average number years working for their current employer was 8 years.

To measure sleep, respondents indicated the average number of hours they sleep per night. These numbers ranged from 4–10 hours per night with an average of 6.65 hours of sleep per night. Prior research has found that adults should regularly sleep seven or more hours of sleep per night in order to support optimal health and that getting less than this number results in a wide variety of negative health consequences (Vincent et al. 2018). Based on this information, the respondents were

classified into two categories. These two categories were those that slept six or less hours per night and those that slept seven or more. There were 134 respondents that fell into the first category (slept 6 or less) and 186 respondents that fell into the second category (slept 7 or more).

All scales used in this study were unidimensional and used a 7-point Likert scale. Grit ($\alpha = 0.74$) was measured using three items from Duckworth and Quinn (2009). Happiness ($\alpha = 0.73$) was measured using three items from Lyubomirsky and Lepper (1999). Perceived Organizational Support (POS, $\alpha = 0.93$) was measured using an eight-item scale from Eisenberger et al. (1986) while Perceived Supervisory Support (PSS, $\alpha = 0.95$) was measured using an eight-item scale from Kottke and Sharafinski (1988). Job Satisfaction ($\alpha = 0.91$) was measured using a three-item scale by Jaramillo, Mulki, and Solomon (2006). Organizational Commitment ($\alpha = 0.91$) was measured using a three-item scale by O'Reilly and Chatman (1986). Finally, Turnover Intentions ($\alpha = 0.92$) was measured using a four-item scales by Brashear, Manolis, and Brooks (2005). Since there was a high degree of reliability for each of the measures, each scale was averaged so that a comparison between means could be conducted.

Findings

A comparison between means was used to examine if differences existed for each scale included in this study based on the two sleep categories previously created. Table 1 displays the results for each scale.

Table 1. Results from comparison between means by sleep.

Scale	Avg. Hours of Sleep per Night	N	Mean	t-statistic	Sig.
Grit	6 or Less Hours	134	6.07	4.79	0.03
	7 + Hours	186	6.26		
Happiness	6 or Less Hours	134	5.10	1.61	0.21
	7 + Hours	186	5.28		
POS	6 or Less Hours	134	5.49	6.21	0.01
	7 + Hours	186	5.10		
PSS	6 or Less Hours	134	5.63	4.60	0.03
	7 + Hours	186	5.29		
Job Sat	6 or Less Hours	134	5.72	4.67	0.03
	7 + Hours	186	5.38		
Org Com	6 or Less Hours	134	5.43	8.20	0.00
	7 + Hours	186	4.91		
Turnover Intentions	6 or Less Hours	134	2.47	5.83	0.02
	7 + Hours	186	2.95		

For Grit, the results indicate a significant difference between those getting 6 or less hours of sleep than those getting 7 or more ($t = 4.79, p = .03$). Employees getting 7 or more hours of sleep ($M = 6.26$) had significantly higher levels of grit than those getting 6 or less hours ($M = 6.07$). For Happiness, the results show no significant differences between the two sleep groups ($t = 1.61, p = .21$). For Perceived Organizational Support (POS, $t = 6.21, p = .01$) and Perceived Supervisory Support (PSS, $t = 4.60, p = .03$), significant differences were found between the two sleep categories. In each case, respondents who slept 6 or less hours had higher perceptions of support than those who slept at least 7 hours per night (POS $M_{6orLess} = 5.49$ vs $M_{7+} = 5.10$; PSS $M_{6orLess} = 5.63$ vs $M_{7+} = 5.29$).

Finally, when examining the job outcomes of job satisfaction, organizational commitment, and turnover intentions, the results indicate that there are significant differences between the two sleep categories for each construct. For job satisfaction ($t = 4.67, p = .03$), employees who experienced less sleep exhibited higher levels of job satisfaction than those that obtained more sleep ($M_{6orLess} = 5.72$ vs $M_{7+} = 5.38$). For organizational commitment, similar results are obtained such that less sleep by employees led to higher levels of organizational commitment ($M_{6orLess} = 5.43$ vs $M_{7+} = 4.91$). In addition, employees who slept six or less hours of sleep per night were less likely to leave the organization than those who slept more ($M_{6orLess} = 2.47$ vs $M_{7+} = 2.95$).

Discussion

This *exploratory study* is the first one to examine if differences in sleep duration affecting two individual and five organizational characteristics for business-to-business salespeople. Results show that differences existed for grit, perceived organizational support, perceived supervisory support, job satisfaction, organizational commitment, and turnover intentions. No difference in happiness levels for business-to-business salespeople was found based on sleep duration.

For the individual characteristics, differences in grit was found based on sleep duration, such that, salespeople sleeping seven or more hours per night were grittier than those that slept less. This is similar to prior research outside of the sales

literature (Siah et al. 2018). This difference is supported based on the conservation of resources (COR) theory (Hobfoll 1989). This theory describes how individuals attempt to build, retain, and guard valuable resources (Mallin and Mayo 2006). These resources are categorized into four types: (1) objects, (2) energies, (3) personal characteristics, and (4) conditions (Mayo and Mallin 2010). In this situation, sleep deprivation is a condition which will affect the amount of grittiness an individual salesperson exudes. Therefore, due to the resource condition of low sleep duration, the salesperson will conserve the personal characteristic of grit. Contrary to prior research (Norlander, Johansson, and Bood 2005; Scott and Judge 2006), there were no significant differences in salesperson happiness between the two sleep categories. However, when examining the prior literature, neither study examined sleep duration which could explain the difference in results.

When examining the organizational characteristics, there were significant differences in the means for all five characteristics based on average hours of sleep per night. As it relates to support, from both the supervisor and organization, salespeople with fewer hours of sleep perceived higher levels of support. Although this may seem counterintuitive, COR theory posits that “when resource loss circumstances are high, resource gains become more important – they gain in value” (Hobfoll et al. 2018, 105). Therefore, the lower average sleep duration by salespeople results in higher perceptions of supervisory and organizational support because the support provided is of higher value to these salespeople.

Similar to support, job satisfaction and organizational commitment were statistically higher for salespeople that slept on average six or less hours. Although this is contrary to the job satisfaction finding of Barnes, Ghumman, and Scott (2013), based on COR theory (Hobfoll 1989), these results can be explained. Again, using COR, the lower average sleep duration by salespeople would result in higher perceptions of job satisfaction. In other words, in a low sleep condition, salespeople are depleted in their resources; therefore, smaller aspects of their job hold more importance to them, leading to higher job satisfaction. These smaller aspects are not considered as valuable by the high sleep salesperson. The current study results also coincide with prior literature where

individuals that perceive higher levels of organizational and supervisory support experience higher job satisfaction (Edmondson and Boyer 2013; Riggie, Edmondson, and Hansen 2009). For organizational commitment, this was the first study to investigate the effect of sleep. Similar results to job satisfaction were found. These results can also be explained by COR (Hobfoll 1989) and prior literature that investigates the relationship between support and organizational commitment (Edmondson and Boyer 2013; Riggie, Edmondson, and Hansen 2009).

Finally, this study found that turnover intentions was lower for salespeople that slept six or less hours per night, on average. Although Tucker, Jimmieson, and Jamieson (2018) found a positive correlation between sleep disturbances and turnover intentions, this study examined sleep duration instead of sleep disturbances. Conservation of resources theory can also be used to explain this finding. Salespeople who experience less sleep are less likely to continue to deplete their resources by searching for other jobs. In other words, when a salesperson does not get adequate sleep, they are more likely to stay with their current employer in order to minimize resource depletion. In addition, this study lines up with what has been found in the support and turnover intention literature stream such that individuals who perceive higher organization and supervisory support experience lower turnover intentions (Edmondson and Boyer 2013; Riggie, Edmondson, and Hansen 2009). Furthermore, this study matches the job satisfaction, organizational commitment, and turnover intentions literature which has found that employees who are more satisfied with their job are more committed to the organization and less likely to leave the firm (Rutherford et al. 2009).

For sales organizations, this study has some interesting implications. Although this research implies that organizations desire sleep deprived employees since this would increase perceptions of support, job satisfaction and organizational commitment while lowering turnover intentions, this is not the recommended course of action. There is a significant amount of prior research which has focused on the negative impact that poor sleep has on both individuals and businesses. Some of these negatives include lost productivity, slower employee decision making, higher absenteeism, higher medical care

costs, increased accident rates, and reduced morale (e.g., Gaultney and Collins-McNeil 2009; Harrison and Horne 1999; Robbins and Jean-Louis 2018; Scott and Judge 2006). Because of this, organizations should not strive for sleep deprived employees. Instead, organizations need to make sure they are providing adequate levels of support to all salespeople so that even those salespeople who obtain seven or more hours of sleep per night feel supported.

The organization needs to encourage salespeople to practice good sleep behaviors by creating good sleep policies. “A good sleep policy is smart business strategy. People think they’re saving time and being more productive by not sleeping, but in fact they are cutting their productivity drastically” (Quoted by Fryer 2006). Some examples of policies and guidelines that sales organizations could incorporate include:

- Encouraging Reasonable Work Hours
- No Texting or Emailing During Typical Sleep Hours
- Taking Travel Time in Consideration when Scheduling Meetings
- Eliminating or Limiting the Usage of Red-Eye Flights
- Creating a Nap Room or Quiet Area in the Office
- Encouraging Exercise and Healthy Living
- Offering Employees Information on How to Create Good Sleep Rituals/Environment at Home
- Limiting the Number of Consecutive Travel Days for Salespeople
- Providing Health Care that Covers the Cost for Sleep Disorders
- Limiting Emotional Exhaustion of Employees
- Giving Employees Regular, Predictable Schedules
- Allowing Employees Flexibility when Creating Schedules
- Allowing for Frequent Breaks to Avoid Employee Decision Fatigue
- Encouraging Work-Life Balance
- Providing Educational Programs on Sleep
- Creating Programs that Incentivize Sleep
- Creating Commitment Devices such as Sleep Pledges or Sleep Challenges
- Switching to Blue-Enriched White Lighting or Natural Light in the Office
- Encouraging the Usage of Sleep Monitoring Apps or Smart Watches
- Providing Training to Supervisors Regarding Sleep and Fatigue Management

By providing salespeople with the necessary resources to obtain better sleep habits, organizations will benefit from more gritty salespeople while also experiencing high levels of happiness, support, job satisfaction, and organizational commitment.

Limitations and future research

As with all academic research, limitations exist that present opportunities for future research. First, this study was exploratory in nature due to the limited amount of research on the subject area, especially in sales. Because of this, additional research is needed which examines the impact that sleep has on sales employees. This research should incorporate more sophisticated models using multi-group analysis. Second, this study focused on sleep duration. However, there are a wide variety of other sleep constructs such as sleep quality, sleep habits, and sleep disturbances that could have been examined. Third, cross-sectional data was used in this study. Because the focus is sleep, it would be worthwhile for a longitudinal study to be conducted examining sleep behaviors over time. Fourth, this research used a nationwide survey of business-to-business salespeople across a variety of industries in order to enhance generalizability. However, additional differences may exist based on the type of sales position, industry, or geographical reach (e.g., global vs. local). Finally, this research focused on grit, happiness, support, job satisfaction, organizational commitment, and turnover intentions. Future research may also want to add additional constructs such as emotional exhaustion, engagement, emotional intelligence, work-family conflict, ethics, etc. to further understand the impact that sleep has on sales employees.

Funding

Funding for this project was provided by the Dean’s Fund for Research Excellence, Jones College of Business.

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